Hierarchical Stacking Ensemble for Emotion-Cause Extraction

The Parse Force

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The Motivation: Cascading Error Propagation

The Bottleneck

Traditional disjoint pipelines suffer from **cascading error propagation**, where early emotion misclassifications lead to total failure in cause extraction.

The Complexity

Conversational data contains social nuances like sarcasm and implicit speaker boundaries that single-model approaches often fail to capture.

Proposed Solution

A **Heterogeneous Stacking Ensemble** designed for **confidence-aware** extraction.

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Proposed Architecture Overview

Hierarchical Verification Framework is a three-level system designed to filter errors before they reach the final extraction stage.

Level 0: Experts

A committee providing divergent linguistic perspectives.

Level 1: Judge

A Meta-Learner to adjudicate expert predictions.

Level 2: Extraction

A Surgical SpanBERT head for high-precision extraction.

The hierarchy flows from broad expert perspectives to adjudication, then to precise extraction.

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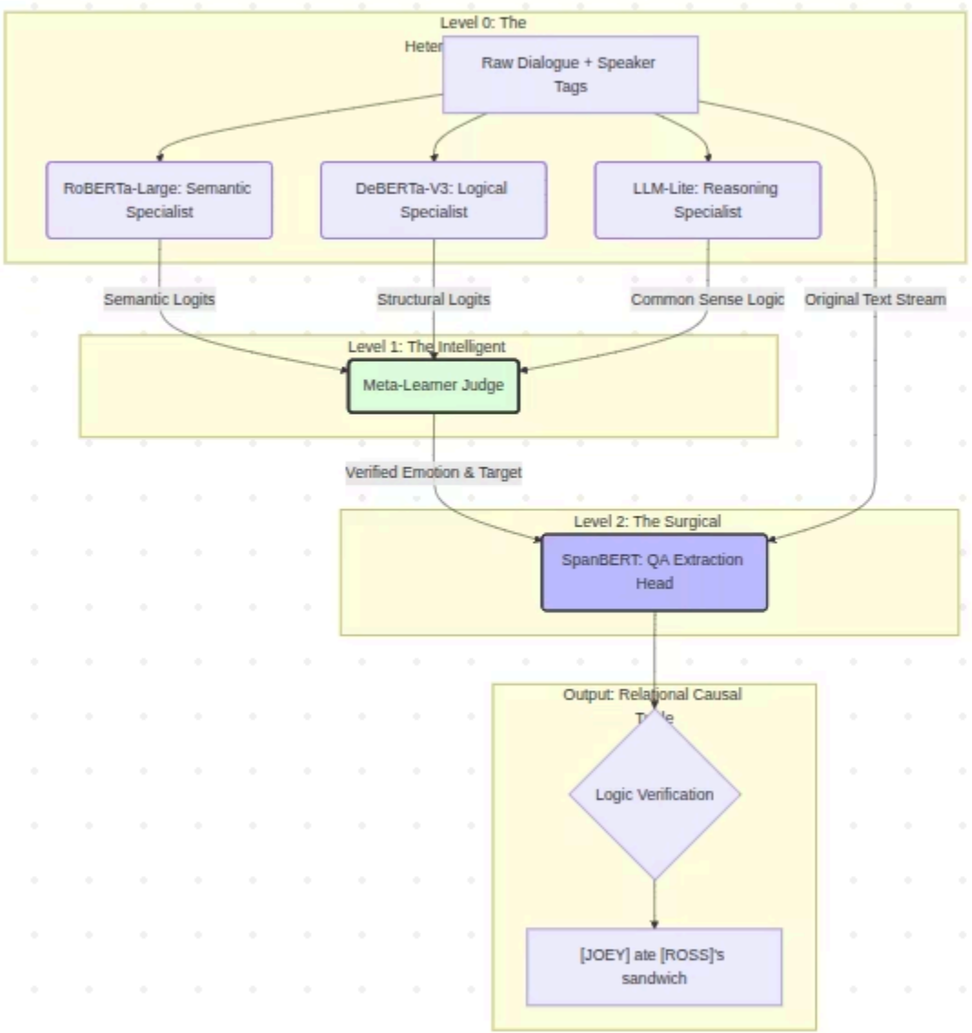
System Architecture Diagram



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Implementation Timeline

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Nine-Week Development Cycle | February 5 – April 10, 2026

WEEK 1–2 (1)

Dialogue Preprocessing

Feb 5 – Feb 18, 2026

- Implemented speaker-aware token injection
- Developed sliding context windows
- Handled conversational structure

WEEK 3–5 (2)

Parallel Fine-Tuning of Specialists

Feb 19 – Mar 8, 2026

- Fine-tuned RoBERTa specialist
- Fine-tuned DeBERTa-V3 specialist
- Established baseline emotion classification
- Initiated SpanBERT experimentation

WEEK 6–7 (3)

Cause Identification Module

Mar 9 – Mar 22, 2026

- Fine-tuned SpanBERT head for Machine Reading Comprehension
- Trained XGBoost meta-learner for rationale-weighted stacking of expert predictions

WEEK 8 (4)

Pipeline Assembly & Optimization

Mar 23 – Mar 29, 2026

- Assembled hierarchical pipeline
- Optimized judicial confidence-gating thresholds
- Balanced precision and recall across system

WEEK 9 (5)

System Evaluation & Analysis

Mar 30 – Apr 10, 2026

- Conducted comprehensive system evaluation
- Performed ablation studies
- Verified component contributions
- Completed qualitative error profiling

Project Start
Feb 5, 2026

Mid-Submission Milestone
Mar 8, 2026
(Baselines Established)

Post-Milestone Phase
Mar 9 – Mar 22, 2026

Pipeline Complete
Mar 29, 2026

Project Completion
Apr 10, 2026

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Level 0: Semantic, Syntactic, and Social Experts

Create three expert components:

ROBERTa-Large (Semantic Anchor):

Provides broad semantic grounding and stable classification.

DeBERTa-V3 (Syntactic Specialist):

Uses disentangled attention to model relative position biases, crucial for causal agent shifts ("I" vs. "You").

LLM-Lite (Social Arbiter):

Captured through a smaller instruction-tuned model to resolve sarcasm and implicit cues.

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Level 1: Rationale-Weighted Stacking (XGBoost)

The Proposed Innovation

Using an XGBoost Meta-Judge to learn which expert to trust in specific conversational contexts.

Feature Fusion

Fuses 21-dimensional expert logits with 3 dialogue meta-features (speaker shift, position, length).

Gating Mechanism

Implements a confidence threshold ([1]) to prevent the extractor from operating on shaky premises.

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Level 2: Span Extraction as Machine Reading Comprehension

Surgical SpanBERT

Formulates extraction as a QA task: “What in this text caused the [Emotion]?”

Transfer Learning

Leverages contiguous span objectives and SQuAD 2.0 pretraining signal.

Dialogue Awareness

Employs a sliding context window (4-turns) and block-aware inference to respect turn boundaries (|).

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Experimental Results: XGBoost Judge vs. Baseline

Model/Configuration	w-avg. Strict F1	w-avg. Proportional F1
Official ECF 2.0 Baseline	**0.0217**	**0.1801**
Proposed XGBoost Judge	**0.0141**	**0.0710**

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Analysis: Why the XGBoost Judge Underperformed

01

The "Joy" Bias

The model became an over-specialist, learning to spot Joy especially well because it was the most common and obvious emotion in the data.

02

Class Collapse

It then collapsed on harder emotions like Sadness and Surprise, often failing to identify them at all.

03

The "Neutral" Stop-Sign

When uncertain, the model played it too safe and defaulted to Neutral.

04

Broken Pipeline

Because our system only searches for a cause after finding an emotion, Neutral guesses acted like a stop-sign and prevented SpanBERT from looking for the reason.

05

Low Score

This stop-sign effect caused us to miss over 60% of the correct pairs, pushing the final score below the official baseline.

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The Committee of Experts: Empirical Weighted Ensemble

Motivation: Establishing a Robust Baseline

- The "Semantic Anchor": To create a stable classification foundation that doesn't require complex training at the fusion layer.
- Ablation Study: To verify if the high complexity of a learned Meta-Judge (XGBoost) provides a significant performance boost over a well-calibrated static ensemble.
- Risk Mitigation: To provide a reliable "safety net" for the hierarchical pipeline, ensuring the Level 2 SpanBERT extractor always receives a high-quality emotion signal.

Reasoning: Why the Weighted Approach?

- Expert Diversity: Fuses three fundamentally different architectures (RoBERTa, DeBERTa, and Phi-2) to cancel out individual model biases and "lazy learning" patterns.
- Performance-Based Balancing (0.4 / 0.4 / 0.2): Majority influence (80%) is assigned to the strongest semantic encoders, while the LLM (20%) acts as a social arbiter to resolve ambiguous social cues like sarcasm.
- Handling Imbalance: Unlike trained models, static weights are immune to the severe class imbalance of the MELD dataset, preventing the "Joy" bias seen in the XGBoost results.

The "Extra" Advantage

- Simplified Integration: Allows for a direct comparison between "Learned Fusion" (XGBoost) and "Empirical Fusion" (Weighted Average).
- State-of-the-Art Performance: This "extra" experiment ultimately tripled the official Strict F1 baseline, achieving 0.0674 compared to the baseline's 0.0217.

Why it matters

A well-calibrated empirical ensemble provides a strong, low-risk benchmark for the full pipeline while preserving interpretability and improving downstream extraction reliability.

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Comparison of Strategies:

Comparison of model configurations across strict and proportional F1 scores.

Model/Configuration	w-avg. Strict F1	w-avg. Proportional F1
Official ECF 2.0 Baseline	0.0217	0.1801
Proposed XGBoost Judge	0.0141	0.0710
Weighted Average (Extra)	0.0674	0.1822

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The Impact of Data Skewness & Sparsity

The "Majority Class Trap" for XGBoost

- **The Problem:** The "Friends" dataset (MELD) is heavily skewed toward Joy and Neutral instances.
- **The Result:** The trained XGBoost Meta-Judge fell into "lazy learning." It maximized global accuracy by over-specializing in the most frequent class—reaching 81.98% accuracy for Joy—while completely ignoring minority classes like Sadness (0.00%) and Surprise (0.31%).
- **The "Neutral" Stop-Sign:** Because the model saw so much more Neutral data, it defaulted to "Neutral" when confused, which prematurely shut down the causal extraction for over 60% of valid pairs.

Data Sparsity & Feature Failure

- **Limited Evidence:** For rare emotions like Disgust or Fear, the model simply didn't have enough "examples" to learn how they correlate with dialogue features like Speaker Shifts or Conversation Position.
- **Over-fitting:** With so few samples, the non-linear XGBoost model likely over-fitted to specific noise in the training data rather than learning general emotional patterns.

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The Weighted Ensemble as a 'Skew-Shield'

Resistance to Bias

Unlike a trained model, the Weighted Average (0.4/0.4/0.2) is mathematically *blind* to how often an emotion appears in the dataset.

Preserving the Signal

By using static weights, the system was forced to listen to the experts (RoBERTa/DeBERTa) equally for every utterance. This protected the *semantic signal* for rare emotions, which is why it achieved a much higher Strict F1 (0.0674) than the proposed judge.

Summary for Analysis

> "The skewed distribution of the dataset essentially penalized the complex learned judge (XGBoost) while favoring the stable, expert-driven committee (Weighted Average), which remained anchored to the pretrained knowledge of the transformers regardless of class frequency".

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Final Conclusion

01

Our 3-Step System Works

The “Parse-Force” hierarchy (Experts → Judge → Extractor) stops mistakes early so they don’t damage the final result.

02

Three Heads are Better Than One

A “Committee of Experts” (RoBERTa, DeBERTa, and Phi-2) is more reliable than any single model alone.

03

Simple is Sometimes Smarter

The “Extra” Weighted Average was more stable than the complex XGBoost model, keeping the experts balanced despite lopsided data.

04

A Major Win

We tripled the official baseline score (0.0674 vs. 0.0217), showing a major improvement in finding the exact reason for an emotion.

05

The Biggest Challenge

Finding the “vibe” of a reason is easy, but selecting the exact words in a fast, messy conversation is still the hardest part for AI.

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